Assignment two

1. Discuss the various networking devices

(a) Router - Directs data packets between different networks and connects devices to the internet

(b) Access points - Allows wireless devices to connect to a wired network, forming a wireless local area network.

(c) Firewall - protects a network by controlling incoming and outgoing network traffic, based on security rules

(d) Hub - A basic device that connects multiple computers, broadcasting data to all devices within a network

(e) Switch - Connects multipe devices within a local area network managing data flow between them

2. Describe the following types of networks

(a) Local Area Network - A LAN connects devices within a small geographic area such as a building or campus.

(b) Wide Area Network - A WAN covers a broad geographic area, such as a city, country, or even globally. It connects multiple LANs over large distances

(c) Metropolitan Area Network (MAN) - A (MAN) spans a city or a large campus. Its larger than a LAN but smaller than a WAN.

3. State and explain the various networking topologies

a) Bus topology -> All devices are connected to a single central cable

b) Ring topology -> Devices are connected in a circular fashion, with each device linked to two others.

c) Star topology - Each device is connected to a central hub or switch

d) Mesh topology - Every device is connected to every other device.

4. Explain the impact of computers to an organisation

(a) Communication - computers facilitate instant communication through various messaging platforms

(b) Cost Reduction - Automation reduces the need for manual labour, cutting operational costs

(c) Security - Computers enable the implementation of data protection mechanisims.

d) Productivity - Automation of repetitive tasks allows employees to focus on more strategic work
e) Global reach. Computers, through the internet, allow organisations to expand their market places

5. Discuss the emerging trends of computers to the society
a) Quantum Computing - An advanced form of computing that performs complex calculations much faster
b) Artificial Intelligence - It enables machines to learn, reason and make decisions, impacting industries like healthcare and transportation.
c) Cloud Computing. This trend allows users to store, manage, and process data remotely, reducing the need for local servers and providing on demand access
d) 5G Technology - The next generation of mobile networks that provide faster internal speeds.
e) Internet of things - It connects everyday devices to the internet enabling automation and smarter living environments.

6. Outline five uses of the internet
a) Communication - Email and video confrencing allow people to connect globally.
b) Online shoping - E commerce platforms enable users to buy goods and services online.
c) Education - E-learning platforms and online cources make education more accesible
d) Entertainment. The internet provides access to to streaming services, games, and social media for entertainment purposes
e) Information sharing - The internet provides access to vast amounts of information through websites, blogs, and social media.